Answer **all** questions in the spaces provided.

1 A golf ball is hit from point A on the ground and moves through the air to point B. The path is illustrated in Fig. 1.1.

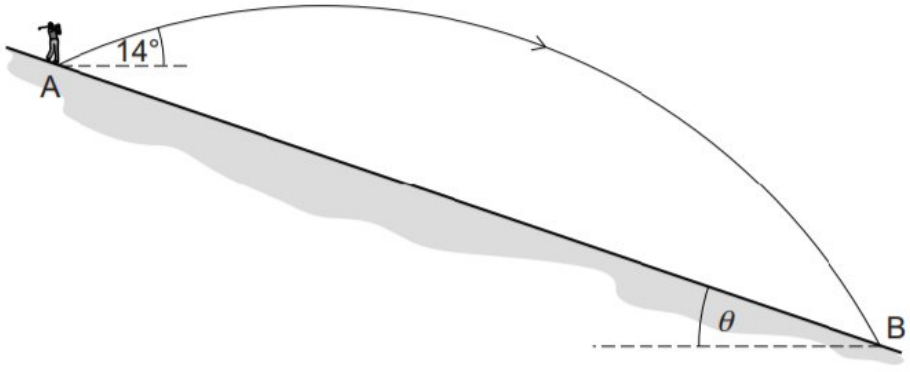


Fig. 1.1

The ground slopes downhill with a constant gradient. The ball has an initial velocity of 67 m s^{-1} at an angle of 14° to the horizontal. The ball hits the ground at B after 4.9 s.

(a) Ignoring air resistance, calculate

(i) the horizontal and vertical components of the ball's velocity at A,

horizontal component at A = ... m s^{-1}

vertical component at A = ... m s^{-1} [1]

(ii) the horizontal displacement from A to B,

horizontal displacement = ... m [2]

(iii) the vertical displacement from A to B,

vertical displacement = ... m [2]

(iv) and the angle of the slope to the horizontal, θ.

angle θ = ... ° [2]

(b) In reality, air resistance acts on the ball in the direction opposite to its motion.

(i) On Fig. 1.1, sketch a likely path of the ball hit from A when air resistance is taken into account. [2]

(ii) Give reasons for the shape you have drawn in **(b)(i)** for the path of the ball at the start.

.....

.....[1]

[Total: 10]